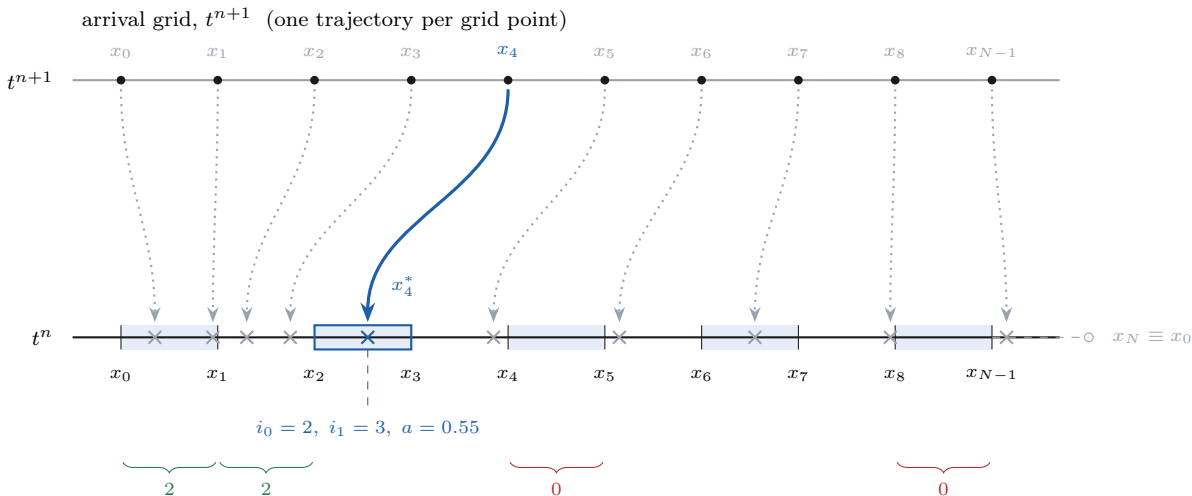




$$f^{n+1}(x_i) = f^n(x_i^*), \quad x_i^* = X(t^n; x_i, t^{n+1}), \quad \frac{dX}{dt} = u(X, t)$$

f is constant along characteristics, so the updated f is an interpolation at the departure point x_i^* , computed from the time-stepped flow map $X(t^n; x_i, t^{n+1})$ determined by the characteristic equation.